FUEL RAIL - 4.0L ENGINE

DESCRIPTION - 4.0L

The fuel rail is mounted to the intake manifold (Fig. 8). It is used to mount the fuel injectors to the engine. On the 4.0L 6–cylinder engine, a **fuel damper** is located near the center of the fuel rail (Fig. 8).

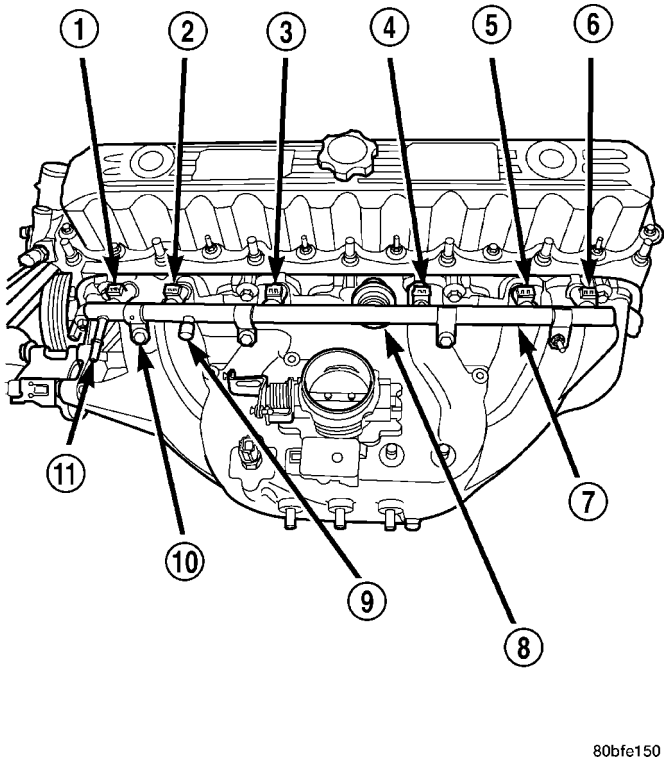

Fig. 8 FUEL RAIL/FUEL DAMPER - 4.0L ENGINE

1 - INJ. #1
2 - INJ. #2
3 - INJ. #3
4 - INJ. #4
5 - INJ. #5
6 - INJ. #6
7 - FUEL INJECTOR RAIL
8 - FUEL DAMPER
9 - PRESSURE TEST PORT CAP
10 - MOUNTING BOLTS (4)
11 - QUICK-CONNECT FITTING

OPERATION - 4.0L

The fuel injector rail supplies the necessary fuel to each individual fuel injector.

The fuel damper is used only to help control fuel pressure pulsations. These pulsations are the result of the firing of the fuel injectors. It is **not used** as a fuel pressure regulator. The fuel pressure regulator is **not mounted** to the fuel rail on any engine. It is located on the fuel tank mounted fuel pump module. Refer to Fuel Filter/Fuel Pressure Regulator for information.

The fuel rail is not repairable.

REMOVAL - 4.0L

The fuel damper is not serviced separately.

WARNING: THE FUEL SYSTEM IS UNDER CONSTANT FUEL PRESSURE EVEN WITH ENGINE OFF. THIS PRESSURE MUST BE RELEASED BEFORE SERVICING FUEL RAIL.

(1) Remove fuel tank filler tube cap.
(2) Perform Fuel System Pressure Release Procedure.
(3) Disconnect negative battery cable from battery.
(4) Remove air tube at top of throttle body. Note: Some engine/vehicles may require removal of air cleaner ducts at throttle body.
(5) Disconnect electrical connectors at all 6 fuel injectors. To remove connector refer to (Fig. 9). Push red colored slider away from injector (1). While pushing slider, depress tab (2) and remove connector (3) from injector. The factory fuel injection wiring harness is numerically tagged (INJ 1, INJ 2, etc.) for injector position identification. If harness is not tagged, note wiring location before removal.
(6) Disconnect fuel supply line latch clip and fuel line at fuel rail. Refer to Quick-Connect Fittings.
(7) Disconnect throttle cable at throttle body. Refer to Throttle Cable Removal/Installation.
(8) Disconnect speed control cable at throttle body (if equipped). Refer to Speed Control Cable in Speed Control System.
(9) Disconnect automatic transmission cable at throttle body (if equipped).
(10) Remove cable routing bracket at intake manifold.
(11) If equipped, remove wiring harnesses at injection rail studs by removing nuts.
(12) Clean dirt/debris from each fuel injector at intake manifold.
(13) Remove fuel rail mounting nuts/bolts (Fig. 10).
(14) Remove fuel rail by gently rocking until all the fuel injectors are out of intake manifold.

INSTALLATION - 4.0L

(1) Clean each injector bore at intake manifold.
(2) Apply a small amount of clean engine oil to each injector o-ring. This will aid in installation.
(3) Position tips of all fuel injectors into the corresponding injector bore in intake manifold. Seat injectors into manifold.
(4) Install and tighten fuel rail mounting bolts to 11 ±3 N·m (100 ±25 in. lbs.) torque.
(5) If equipped, connect wiring harnesses to injection rail studs.
(6) Connect electrical connectors at all fuel injectors. To install connector, refer to (Fig. 9). Push con-